luca.secchieri@gmail.com
+39 331 8646074

Luca Secchieri

linkedin.com/in/luca-secchieri
github.com/Liuck27

Software engineer with experience building production-grade systems spanning AI pipelines, backend services, and developer tooling. Proven track record delivering high-impact solutions, including LLM-based workflows and CI/CD infrastructure, with measurable improvements in efficiency and reliability. Background in safety-critical avionics software brings a strong engineering rigor to high-performance systems.

Skills

- Programming Languages: Python | C | C++
- AI: RAG | LLM Agents | Prompt Engineering and Evaluation | Ollama | MCP
- Backend & WebDev: FastAPI | REST
- Databases: PostgreSQL |SQLite | ChromaDB
- Systems & Infrastructure: Linux | Docker | AWS (EC2, S3, Lambda, IAM)
- DevOps: Git | GitHub Actions | GitLab CI
- Monitoring: Grafana | Prometheus

Work Experience

Software Engineer — **Leonardo S.p.A.** — **Dec 2024–Present**
Leonardo Helicopters Division — Milan, Italy

- **Reduced manual review time by 95%** by building a vision LLM pipeline to detect semantic content differences, iteratively refining prompts and evaluating model accuracy across curated test sets to maximize detection reliability.
- **Reduced integration time by 80%** by designing and deploying the division's first CI/CD pipeline (GitLab CI), automating builds, static analysis, and testing.
- Streamlined development workflows by engineering a **local agentic RAG system** that integrated software requirements retrieval with LLMs, enabling engineers to locate and reference specifications during development.
- Improved delivery predictability by building a full-stack dashboard (React, Python) that tracked GitLab issue progress via daily snapshots and forecasting models.
- Led integration of contributions from 10+ engineers into production releases while developing and verifying embedded software under DO-178C DAL A constraints, producing traceable artifacts for avionics certification.

Projects

Event-Driven Quantitative Backtesting Framework — **https://quant-backtester**
Python · FastAPI · PostgreSQL · React · Docker

- Designed a production-grade event-driven backtesting engine with realistic slippage and commission modeling.
- Built an async REST API with SSE streaming for real-time equity curve updates and live parameter sweep progress.
- Deployed a full-stack application across Vercel (React), Render (Dockerized API), and Supabase (PostgreSQL).

MLOps Fraud Detection Platform — **github.com/Liuck27/ml-fraud-detection-platform**
Python · FastAPI · PostgreSQL · Docker · Airflow · MLflow · XGBoost · PyTorch · Prometheus

- Served two models (XGBoost + PyTorch autoencoder) behind a FastAPI inference API with deterministic MD5-hash A/B routing on 284,807 transactions, returning SHAP TreeExplainer top-k contributions on every /predict call.
- Shipped a 6-service Docker Compose stack (Airflow, MLflow, FastAPI, Postgres, Prometheus, Grafana) with 4 custom Prometheus metrics, a 4-panel Grafana dashboard, and an MLflow registry gated on PR-AUC for champion promotion.

Education

- **Control Systems Engineering - Master's Degree,** University of Padua — **2024**
- **Robotics and Automation Engineering - Erasmus Program,** Universitat Politecnica de Catalunya — **2024**
- **Information Engineering - Bachelor's Degree,** University of Padua — **2022**

Certifications

- **Oracle Cloud Infrastructure** 2025 Certified Generative AI Professional
- Machine Learning for Financial Trading - New York Institute of Finance